

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION with OpenCV

COURSE CODE: DSA02

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CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

1. Grey Level Transformation

(a) Grey level transformation remaps every pixel intensity through a function $s = T(r)$ so that the visual contrast and appearance of an image can be enhanced. It is used to brighten/darken images, expand contrast in a chosen range, or reverse intensities to highlight detail hidden in bright or dark regions, without changing the spatial arrangement of pixels.

(b) Negative transformation $s = 255 - r$:

$$r = 50 \rightarrow s = 255 - 50 = 205$$

$$r = 100 \rightarrow s = 255 - 100 = 155$$

$$r = 150 \rightarrow s = 255 - 150 = 105$$

$$r = 200 \rightarrow s = 255 - 200 = 55$$

2. Log Transformation

(a) Log transformation, $s = c \cdot \log(1+r)$, compresses the higher end of the intensity range while stretching the lower (dark) end, because the log function grows quickly for small r and slowly for large r . This brings out details in dark regions of an image (e.g., Fourier spectra) that would otherwise be indistinguishable, at the cost of compressing bright details.

(b) $s = 10 \cdot \log_{10}(1+r)$:

$$r = 0 \rightarrow s = 10 \cdot \log_{10}(1) = 10 \times 0 = 0.000$$

$$r = 10 \rightarrow s = 10 \cdot \log_{10}(11) = 10 \times 1.0414 = 10.414$$

$$r = 50 \rightarrow s = 10 \cdot \log_{10}(51) = 10 \times 1.7076 = 17.076$$

$$r = 100 \rightarrow s = 10 \cdot \log_{10}(101) = 10 \times 2.0043 = 20.043$$

3. Histogram Processing

(a) Histogram equalization redistributes the intensity levels of an image so that the output histogram is as close to uniform as possible. This spreads out the most frequent intensity values, increasing the global contrast of the image, especially useful when the original histogram is concentrated in a narrow range.

(b) 3-bit image, total pixels $N = 2+2+4+8 = 16$. PDF and CDF:

r : freq : PDF ($f/16$) : CDF

$$0 : 2 : 0.125 : 0.125$$

$$1 : 2 : 0.125 : 0.250$$

$$2 : 4 : 0.250 : 0.500$$

$$3 : 8 : 0.500 : 1.000$$

Equalized level: $s = \text{round}(\text{CDF} \times (L-1))$, $L = 8 \rightarrow L-1 = 7$

$s_0 = \text{round}(0.125 \times 7) = \text{round}(0.875) = 1$
 $s_1 = \text{round}(0.250 \times 7) = \text{round}(1.75) = 2$
 $s_2 = \text{round}(0.500 \times 7) = \text{round}(3.5) = 4$
 $s_3 = \text{round}(1.000 \times 7) = \text{round}(7.0) = 7$

4. Spatial Filtering (Smoothing)

(a) Spatial smoothing filters replace each pixel with an average (or weighted average) of its neighbourhood. This blurs the image, suppressing high-frequency noise and small variations, and is typically used as a pre-processing step before edge detection or segmentation.

(b) 3×3 averaging filter on the region:

10 20 30
 20 30 40
 30 40 50

Sum = $10+20+30+20+30+40+30+40+50 = 270$

New centre value = $270 / 9 = 30$

5. Spatial Filtering (Sharpening)

(a) Sharpening filters emphasise rapid intensity changes (edges and fine detail) by responding to the second derivative of the image. Adding a Laplacian-type response back to the image boosts edges while leaving flat regions largely unaffected, making the image appear crisper.

(b) Laplacian mask $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$ on:

10 10 10
 10 20 10
 10 10 10

Output = $4 \times (\text{centre}) - (\text{sum of 4 neighbours})$
 $= 4 \times 20 - (10+10+10+10) = 80 - 40 = 40$

6. Frequency Domain Filtering

(a) Low-pass filtering in the frequency domain retains low-frequency components (which carry the bulk of image energy/smooth regions) and attenuates high frequencies, producing a smoothed image. High-pass filtering does the opposite — it removes/attenuates low frequencies and retains high frequencies, sharpening edges and fine detail while removing slowly varying background.

(b) $F = [100, 80, 40, 20]$; low-pass filter keeps only the first two (lowest-frequency) components:

7. Thresholding

(a) Global thresholding converts a grayscale image into a binary image by comparing every pixel to a single fixed threshold T : pixels with intensity $\geq T$ are set to 1 (object/foreground) and pixels below T are set to 0 (background). It is the simplest form of image segmentation.

(b) Pixel value = 200, $T = 100$. Since $200 \geq 100$:

8. Edge Detection

(a) Edge detection locates points in an image where intensity changes sharply, corresponding to object boundaries, texture changes, or discontinuities. It is typically performed by convolving the image with gradient operators (e.g. Sobel, Prewitt) and identifying pixels with large gradient magnitude.

(b) Horizontal Sobel mask $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$ on:

10 10 10

20 20 20

30 30 30

$$\begin{aligned}\text{Gradient } G_x &= (-1 \times 10 - 2 \times 10 - 1 \times 10) + (0) + (1 \times 30 + 2 \times 30 + 1 \times 30) \\ &= -40 + 120 = 80\end{aligned}$$

9. Region-Based Segmentation

(a) Region growing starts from one or more seed pixels and iteratively adds neighbouring pixels to the region as long as they satisfy a similarity criterion (e.g. $|\text{intensity} - \text{seed}| \leq \text{threshold}$). The process stops when no more neighbouring pixels satisfy the criterion, producing a connected region of similar pixels.

(b) Seed = 50, threshold difference = 5 $\Rightarrow$ acceptable range = [45, 55]. Example neighbourhood {48, 52, 60}:

$$|48 - 50| = 2 \leq 5 \rightarrow \text{included in region}$$

$$|52 - 50| = 2 \leq 5 \rightarrow \text{included in region}$$

$$|60 - 50| = 10 > 5 \rightarrow \text{excluded from region}$$

10. Combined Enhancement & Segmentation

(a) Smoothing is applied before segmentation to suppress noise and small spurious intensity fluctuations. Without smoothing, noisy pixels can be mis-classified during thresholding/region growing, producing a fragmented or noisy segmentation mask; smoothing first gives a cleaner, more reliable boundary.

(b) Step 1 — Averaging: given mean = 48.

(c) (No separate part-c; combined with threshold step below.)

(d) Step 2 — Threshold at $T = 50$: since $48 < 50$, the pixel is classified as background (0).